

ECSE 2500

Lec 22

April 20

Topic: Conditional expectation

— Law of iterated expectation

Sum/Average of several RVs

— Law of large number

## □ Conditional expectation

Recall the definition in the last class

Q: What is the expected value of the conditional expectation of  $Y$  given  $X$  (e.g. a function of RV  $X$ ) ?

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

$$= \int_{-\infty}^{+\infty} E[Y|X=x] f_X(x) dx$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y \underbrace{f_{Y|X}(y|x)}_{f_{Y|X}(y|x)} dy \underbrace{f_X(x)}_{f_X(x)} dx$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y \underbrace{f_{XY}(x,y)}_{f_{XY}(x,y)} dx dy$$

$$= \int_{-\infty}^{\infty} y \cdot f_Y(y) dy$$

$$= E[Y]$$

To summarize, we have

$$E[E[Y|X]] = E[Y]$$

Reduce to the simplified  
case of  $h(Y) = Y$ .

The law of iterated expectation!!

Composition/Nest of two Expectations

Slight generalization:

If we have a function  $h(\cdot)$  of a RV  $Y$ ,

$$E[E[h(Y)|X]] = E[h(Y)]$$

That is, we can think of  $E[h(Y)]$  in two steps:  
First, compute  $E[h(Y)|x]$  for each possible  
value  $x$  of RV  $X$ . Secondly, we average over  
the distributions of  $X$ .

### Example

(Discrete RVs) Consider the coin toss example in the last week. Flip 3 coins. and let

$X = \# \text{ of heads}$

$Y = \text{position of first head}$

In the last week, we have computed  $P_{Y|X}$  as

| $P(Y X)$ |   | $Y$ |               |               |               |
|----------|---|-----|---------------|---------------|---------------|
|          |   | 0   | 1             | 2             | 3             |
| $X$      | 0 | 1   | 0             | 0             | 0             |
|          | 1 | 0   | $\frac{1}{3}$ | $\frac{1}{3}$ | $\frac{1}{3}$ |
|          | 2 | 0   | $\frac{2}{3}$ | $\frac{1}{3}$ | 0             |
|          | 3 | 0   | 1             | 0             | 0             |

Given the conditional PMF in the table, we calculate

$$E[Y|X=0] = \sum_{y_j} y_j P(y_j | X=0)$$

$$= 1 \cdot 0 + 1 \cdot 0 + 2 \cdot 0 + 3 \cdot 0 = 0$$

$$E[Y|X=1] = 0 \cdot 0 + (1+2+3) \cdot \frac{1}{3} = 2$$

$$E[Y|X=2] = 0 \cdot 0 + \frac{2}{3} + 2 \cdot \frac{1}{3} + 3 \cdot 0 = \frac{4}{3}$$

$$E[Y|X=3] = 1 \cdot 1 = 1$$

Then we use the law of iterated expectation, we have

$$E[Y] = E[E[Y|X]]$$

$$= E[Y|X=0]P(X=0) + E[Y|X=1]P(X=1) \\ + E[Y|X=2]P(X=2) + E[Y|X=3]P(X=3)$$

$$= 0 \cdot \frac{1}{8} + 2 \cdot \frac{2}{8} + \frac{4}{3} \cdot \frac{3}{8} + 1 \cdot \frac{1}{8} = \frac{11}{8}$$

To do a sanity check,

$$E[Y] = \sum_{y_i} y_i P(Y=y_i)$$

$$= 0 \cdot \frac{1}{8} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} = \frac{11}{8}$$

→ The law of iterated expectation is verified!

**Example**

Consider  $X$  is uniformly drawn from  $[0, 1]$ ,  
(Continuous RVs) and  $Y$  is uniformly drawn from  $[0, x]$ .

Q: What is the expected value of  $Y$ ?

The problem structure facilitates us to use the law of iterated expectation!

① What is  $E[Y|X]$ ?

$$\begin{aligned} E[Y|X] &= \int_0^x y \underbrace{f_{Y|X}(y|x)}_{\frac{1}{x}} dy \\ &= \int_0^x \frac{y}{x} dy \\ &= \frac{1}{x} \frac{y^2}{2} \bigg|_{y=0}^{y=x} = \frac{x}{2} \end{aligned}$$

② What is  $E[Y]$ ?

$$\begin{aligned} E[Y] &= E[E[Y|X]] \\ &= \int_0^1 \frac{x}{2} \cdot f_X(x) dx \\ &= \int_0^1 \frac{x}{2} dx = \frac{x^2}{4} \bigg|_{x=0}^{x=1} = \frac{1}{4} \end{aligned}$$

To do a sanity check, from previous notes, we have calculated  $f_Y(y) = -\log y$ , and then

$$E[Y] = \int_0^1 y \log y dy = \frac{y^2}{4} - \frac{y^2}{2} \log y \bigg|_{y=0}^{y=1} = \frac{1}{4}$$

## □ Sum/Mean of several random variables

In many situations, we observe a sequence of RVs (e.g., a series of coin flips) and want to process these to extract information from those RVs.

A natural way to do so is via the sum/mean of RVs.

$$\text{Sum: } S_n = X_1 + X_2 + \dots + X_n = \sum_{i=1}^n X_i$$

↑  
number of RVs

$$\text{Mean: } M_n = \frac{1}{n} S_n = \frac{1}{n} \sum_{i=1}^n X_i$$

First we compute the expected value of these  $S_n$  and  $M_n$

$$\begin{aligned} E[S_n] &= E[X_1 + X_2 + \dots + X_n] \\ &= \sum_{i=1}^n E[X_i] \end{aligned}$$

$$\begin{aligned} E[M_n] &= \frac{1}{n} E[X_1 + X_2 + \dots + X_n] \\ &= \frac{1}{n} \sum_{i=1}^n E[X_i] \end{aligned}$$

That is, the expected value of Sum of RVs is sum of Expected RVs.

Second, what about their variance?

$$\begin{aligned}\text{Var}(S_n) &= E[(S_n - E[S_n])^2] \\&= E[S_n^2] - (E[S_n])^2 \\&= E[(X_1 + X_2 + \dots + X_n)(X_1 + X_2 + \dots + X_n)] \\&\quad - (E[X_1 + X_2 + \dots + X_n])^2 \\&= E\left[\sum_{i=1}^n \sum_{j=1}^n X_i X_j\right] - \sum_{i=1}^n \sum_{j=1}^n E[X_i] E[X_j] \\&= \sum_{j=1}^n [E[X_j^2] - (E[X_j])^2] \\&\quad + \sum_{i=1}^n \sum_{j \neq i} [E[X_i X_j] - E[X_i] E[X_j]] \\&\quad \quad \quad = (X_i - E[X_i])(X_j - E[X_j]) \\&= \sum_{j=1}^n \text{Var}(X_j) + \sum_{i=1}^n \sum_{j \neq i} \text{Cov}(X_i, X_j)\end{aligned}$$

That is, the variance of sum of RVs is the sum of all the possible covariances.

Special cases

- 1)  $\text{Cov}(X_i, X_j) = 0$ , for all  $i \neq j$   
all uncorrelated

$$\text{Var}(S_n) = \sum_{j=1}^n \text{Var}(X_j)$$

2) All RVs  $X_i$  are independent and identically distributed (iid). That is, each  $X_i$  has the same PDF  $f_X(x)$  and their covariance are zero,

(iid)  $\xrightarrow{\text{implication}}$  
$$\begin{cases} E[X_i] = \mu, & \text{for all } i \\ \text{Var}(X_i) = \sigma^2, & \text{for all } i \\ \text{Cov}(X_i, X_j) = 0, & \text{for } i \neq j \end{cases}$$

Sum  $S_n$  
$$\begin{cases} E[S_n] = \sum_{i=1}^n E[X_i] = n\mu \\ \text{Var}(S_n) = n\sigma^2 \end{cases}$$

Mean  $M_n$  
$$\begin{cases} E[M_n] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \mu \\ \text{Var}(M_n) = \frac{1}{n^2} \text{Var}(S_n) = \frac{\sigma^2}{n} \quad (*) \end{cases}$$

$\uparrow$   
 $\text{Var}(aX) = a^2 \text{Var}(X)$

The equation (\*) is important since it says that the variance of  $M_n$  gets smaller as the number of RVs increases!!

In the limit, for  $n$  iid. RVs,

$$\lim_{n \rightarrow \infty} \text{Var}(M_n) = 0,$$

$$\lim_{n \rightarrow \infty} E[M_n] = \mu.$$

So we get closer and closer to the right mean of  $X$ : when  $n \rightarrow \infty$ .